Journal of CO_2 Utilization 37 (2020) 158–172

Contents lists available at ScienceDirect

Journal of CO_2 Utilization

journal homepage: www.elsevier.com/locate/jcou

Application of nanoindentation technology for characterizing the mechanical properties of shale before and after supercritical CO_2 fluid treatment

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ARTICLE INFO

Keywords:
Supercritical carbon dioxide
Nanoindentation
Shale gas
Fracture toughness
Continuous stiffness measurement

ABSTRACT

To better implement super critical carbon dioxide (SC-CO_2) fracturing on shale, it is important to investigate effects of SC-CO_2 on rock mechanical properties. Because micro-scale mechanical properties determine the overall mechanical response, the determination of micro-scale mechanical properties is necessary. In this paper, continuous stiffness measurement (CSM) and constant loading rate (CLR) mode nanoindentation was performed on shale samples before and after SC-CO_2 treatment. With the aid of scanning electron microscope and energy dispersive spectra mapping technique, mechanical strength difference on clay-rich and quartz –rich areas in shale sample can be identified. The results demonstrated that samples treated by SC-CO_2 present lower Hardness, Young's modulus and fracture toughness. The average decrease rates of hardness and Young's modulus under CSM mode for three sample are 32.01 % and 11.75 %. The hardness, Young's modulus and fracture toughness under CLR mode for the three samples have decreased at average rates of 29.5 %, 11.0 %, and 11.3 %, respectively. Moreover, the mechanical strength substantially decreases in the clay-rich area because CO_2 adsorption and mineral dissolution happen after the SC-CO_2 treatment. The change in mechanical properties in quartz-rich area is much smaller because there is little effect on quartz. The close examination of indentation fracture geometry indicate more micro fractures have been generated in the post SC-CO_2 treated shale, thus we can expect the capability of SC-CO_2 for rock weaken. This study can provide more insights for fracture mechanism in target areas with different mineral distribution, which is beneficial for SC-CO_2 engineering applications.

1. Introduction

In recent years, shale gas reservoirs have been aggressively unlocked by the advancement of multistage fracturing and horizontal well drilling. Conventional water-based fracturing fluids have been widely used for shale gas development because of the economic and investment balance. However, because of the high clay contents of shale, water can potentially induce clay swelling and pore throat blocking, thereby affecting the natural gas flow velocity. Moreover, the vast consumption of water poses environmental and water supply risks to local life, which has prompted some European countries and several states in the United States of America to make strict regulations on massive hydraulic fracturing treatments [1–3].

Fortunately, these problems can be effectively avoided through using non-aqueous fluids as an alternative. Supercritical carbon dioxide (SC-CO_2) is a special fluid that has unique properties different from those of gaseous CO_2 with a low viscosity and liquid CO_2 with a high density. When the pressure exceeds 7.38 MPa and the temperature exceeds 31.1 °C, CO_2 can transform from a liquid state to a supercritical state [4,5]. There are clear advantages of SC-CO_2 fracturing techniques due to its potential to improve hydraulic fracturing effects, enhance shale gas recovery, and reduce water consumption. Additionally, the

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https://doi.org/10.1016/j.jcou.2019.11.022
Received 9 March 2019; Received in revised form 12 October 2019; Accepted 19 November 2019
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injection of a large volume of CO_2 can achieve greenhouse gas sequestration in low-permeability shale, which is beneficial for decreasing environmental disruption. Therefore, the SC-CO_2 fracturing technique has been considered to be a promising and ideal non-aqueous fracturing method for shale gas extraction and CO_2 storage [6–8].

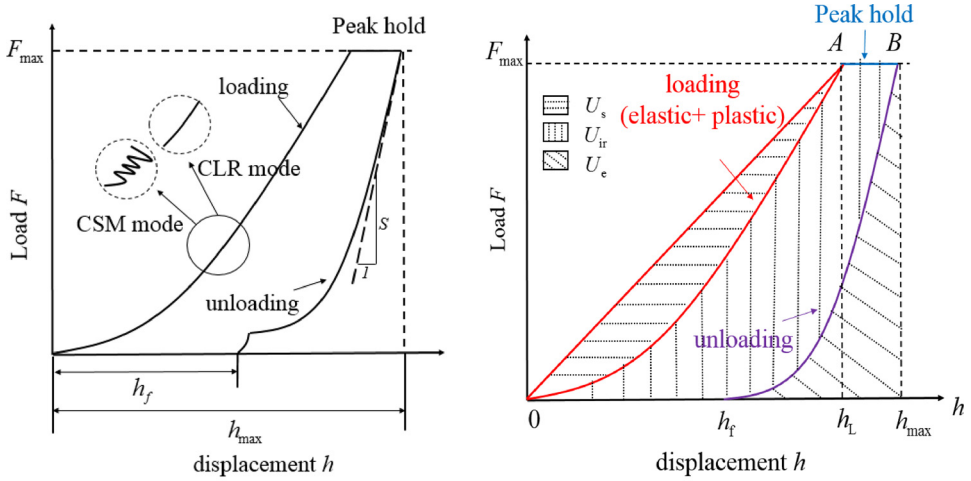
As a multiphase and multicomponent sedimentary rock, shale's elastic properties, such as the elastic modulus, hardness, and fracture toughness, are critical parameters for hydraulic fracturing, wellbore drilling, well completion design and post-hydraulic fracturing analysis. Some researchers have performed a series of tri-axial and uniaxial experiments to observe the mechanical properties changes before and after SC-CO_2 immersion on different rocks, such as shale, sandstone, coal and chalk, but most studies were performed on inch-sized cores [9–12]. Understanding shale mechanical properties is a challenging and intractable task through the use of conventional core-size testing. First, coring of shale is difficult because of the unstable physical and chemical characteristics of shale. Second, conventional mechanical tests are time-consuming and expensive, and the experimental procedure is not repeatable. Third, there is a need for a good method to study the effects of fluid seepage from the surface into interior to investigate the mechanical properties changes in areas with diverse mineral distributions before and after SC-CO_2 treatment.

In recent years, the nanoindentation method has received increasing attention and has been proposed as a good tool to replace conventional mechanical tests. Unlike conventional compression tests, the nanoindentation test depends on an indenter with special mechanical and geometrical properties, and load vs. displacement response curve during loading-unloading process is obtained, which can infer mechanical parameters. Although the nano/micro-scale rock mechanical properties of sandstone, marble, limestone and shale have previously been analyzed, the majority of studies focused on dry rock [13–15]. The working fluid can interact with different rock minerals and induce a series of changes in the mineral and pore structures and mechanical properties, which is meaningful in the understanding of engineering performance [16–18]. According to nanoindentation results of shales treated by fracturing fluid, mineral reactions caused by fracturing fluid are the primary cause for the reduction of the Young's modulus, which has since brought considerable attention to shale-fluid interactions [19]. Except for shale, some authors discussed the geomechanical changes of limestone due to SC-CO_2 injection. These researchers found that SC-CO_2 has two contrasting effects on rock mechanical properties in different areas—well-consolidated areas are weakened, but weaker areas are strengthened [20]. Authors also characterized the changes in the mechanical properties of shale samples from octadecyltrimethoxysilane (OTMS) and discussed the function of surface treatment in shale-softening mitigation. The majority of former studies performed nanonindentation on shale mechanical properties using a quasi-static constant loading rate (CLR) mode, which means that only the mechanical properties of the rock surface below the maximum depth could be investigated. In fact, the seepage of SC-CO_2 into rock can cause different types of damage to the rock surface and the rock interior, and it is beneficial to gain insight into the degradation extent in multi-layered shales [21]. To offset this limitation, continuous stiffness measurement (CSM) is presented for use in mechanical properties measurements throughout the sample thickness, in which a small harmonic force is added on the indenter, and the harmonic response of the indenter is measured at the excitation frequency. The CSM nanoindentation method has also been used to study advanced mechanical properties such as the local strain-rate sensitivity and creep stress exponent [22–24]. However, little work has focused on examining the mechanical properties of shale with CSM nanoindentation, and none has investigated shale mechanical properties before and after SC-CO_2 treatment.

In this paper, we investigate the interaction between shale and SC-CO_2 by nanonindentaion technique. Moreover, scanning electron microscope (SEM) and energy dispersive spectra (EDS) mapping were also combined to observe the mineral and pore structure changes and the fracture morphology of quartz-rich regions and clay-rich regions before and after SC-CO_2 treatment. The CSM and CLR nanoindentation modes were performed on identical shale samples, aiming to not only investigate the effect of SC-CO_2 on the rock surface but also to characterize the interior of the rock. Moreover, the load vs. displacement curve and critical mechanical parameters such as the Young's modulus, hardness and fracture toughness values, were compared before and after SC-CO_2 treatment for quartz-rich regions and clay-rich regions. The rock mechanical properties changes can provide insight into the fracture mechanism and proppant embedment problems in SC-CO_2 fracturing and SC-CO_2 injection for CO_2 storage.

2. Nanoindentation methods

The nanoindentation method is widely used as a surface testing method for the characterization of mechanical properties for materials. A single nanoindentation experiment involves loading of the sample (by applying a user-specified force or specifying depth of investigation into the sample) and subsequently unloading the sample [25,26]. According to the purpose of the study, nanoindentation can be carried out by the CLR (constant loading rate) or the CSM (constant loading strain rate) model. In this study, CLR mode and CSM mode test scenarios were used simultaneously on identical sample. The schematic illustration of load



(a) Schematic illustration of typical load vs. displacement curve (b) Energy analysis for fracture toughness determination

Fig. 1. Nanoindentation schematic diagrams: corresponding load-displacement curve.

vs. displacement curve in nanoindentation is shown in Fig. 1.

2.1. Constant loading rate measurement

The theoretical basis for this approach was established by Joslin and Oliver, and the method allows extracting much more information from a single experiment than does quasi-static nanoindentation [27]. The hardness is usually defined from the project area and the peak load:

$$H = \frac{F_{\max}}{A_c} \tag{1}$$

where A_c is the contact area and $F_{\max}$ is the peak load.

The contact area can be computed by:

$$A_c = 24.5h_c^2 \tag{2}$$

where h_c is the contact depth. To the contact depth can be described as:

$$h_c = h_{\max} - 0.75 \times \frac{F_{\max}}{S} \tag{3}$$

$h_{\max}$ is the maximum indentation depth and S is the contact stiffness, which is calculated by the initial stages of the unloading curve.

$$S = \frac{dF}{dh} \tag{4}$$

Where F is load;

The reduced Young's modulus can be calculated as

$$E_r = \frac{\sqrt{\pi}S}{2\beta\sqrt{A_c}} \tag{5}$$

where β is a constant dependent on the geometry of the indenter (1.034 for a Berkovich indenter)

Technically, the elastic moduli of the samples are calculated by

$$\frac{1}{E_r} = \frac{1-v^2}{E} + \frac{1-v_i^2}{E_i} \tag{6}$$

where E is the Young's modulus of the samples and v is the Poisson's ratio of the samples. E_i and v_i are the Young's modulus and Poisson's ratio of the diamond indenter, respectively. For the Berkovich indenter, $E_i = 1141$ GPa and $v_i = 0.07$.

The fracture toughness represents the resistance offered by a rock material to the growth of a crack. The energy method is usually used to calculate the fracture toughness, and the fracture energy is accounted for as a portion of the irreversible energy instead of the total energy [28]. The elastic energy Ue can be defined as the difference between the total energy U_t and the irreversible energy U_p. U_p is composed of two parts: the pure plastic energy U_{pp} and the fracture energy U_{crack}. In Fig. 5(b), the projection area $0{\cdot}A{\cdot}B{\cdot}h_f$ was defined as the irreversible energy producing with loading, whereas the projection area $h_f{\cdot}B{\cdot}h_{\max}$ was defined as the elastic energy which was recovered after unloading.

The equation of the total energy can be illustrated as

$$U_t = \int_0^{h_m} Fdh = \frac{F_m h_m}{3} \tag{7}$$

U_e can be calculated from unloading curve and defined as:

$$U_e = \int_{h_f}^{h_{\max}} Fdh \tag{8}$$

The relationship between U_t and U_{pp} can be expressed as

$$\frac{U_{pp}}{U_t} = 1 - \left[\frac{1 - 3(\frac{h_f}{h_{\max}})^2 + 2(\frac{h_f}{h_{\max}})^3}{1 - (\frac{h_f}{h_{\max}})^2}\right] \tag{9}$$

Then, the fracture energy U_{crack} can be determined from

$$U_{crack} = U_p - U_{pp} \tag{10}$$

The critical energy release rate G_c can be obtained by

$$G_c = \frac{\partial U_{crack}}{\partial A} = \frac{U_{crack}}{A_m} \tag{11}$$

The fracture toughness K_c can be expressed as

$$K_c = \sqrt{G_c E_r} \tag{12}$$

2.2. Constant loading rate measurement

CSM depends on imposing fast oscillation with high frequency onto the quasi-static loading signal; thus, a harmonic force can be added onto the load. The main advantage of CSM is that it offers direct measurement of the dynamic contact stiffness, S, at any point along the loading curve. Furthermore, CSM is insensitive to thermal drift, and the mechanical results are not considerably influenced by drift errors. Dynamic measurements have been widely used for investigating variations in hardness and Young's modulus within a certain distance from the surface, which is of great interest concerning the interior of a sample. A second-order equation can be used to express the correlation between the response and the force [29]:

$$m\ddot{z}(t) + D\dot{z}(t) + Kz(t) = F(t) \tag{13}$$

Taking the first and second derivatives of $z(t)$ yields

$$\begin{aligned} \dot{z}(t) &= z_0 iwe^{i(wt-f)} \\ \ddot{z}(t) &= -z_0 w^2 e^{i(wt-f)} \end{aligned} \tag{14}$$

where z_0 is the amplitude of the indenter oscillation, w is the oscillation frequency, and ϕ is angle by which the response function $z(t)$ lags the force function $F(t)$.

The contact stiffness for the CSM model is fundamental for calculating the mechanical properties and is a function of indentation depth, which can be computed as

$$S = \left[\frac{1}{(F_0/z_0)\cos\phi - (K_s - Mw^2)} - \frac{1}{K_f}\right]^{-1} \tag{15}$$

where K_f is the stiffness of the indenter frame. M is the mass, and F_0 is the amplitude of $F(t)$.

After obtaining the contact stiffness, the calculation of Young's modulus and hardness can be conducted using Eqs. (3)–(6). Since the Young's modulus and hardness are functions of depth rather than static values under the CSM mode, the stable Young's modulus and hardness can be used to present the bulk mechanical properties for each indentation. Because there is no method for elastic and plastic recovery calculation using the load displacement curve under the CSM mode, fracture toughness cannot be determined.

For simplicity, the averaging method is used to calculate the average mechanical value of the Young's modulus, hardness and fracture toughness values through numerous valid tests on one identical sample. A large number of nanoindentations can provide sufficient data for the bulk properties of rock:

$$\overline{E} = \frac{1}{n}\sum_{j=1}^{n} E \tag{16}$$

$$\overline{H} = \frac{1}{n}\sum_{j=1}^{n} H \tag{17}$$

$$\overline{K_c} = \frac{1}{n}\sum_{j=1}^{n} K_c \tag{18}$$

where $\overline{E}$, $\overline{H}$, and $\overline{K_c}$ are the average Young's modulus, hardness, and fracture toughness values, respectively.

3. Experimental procedure

3.1. Sample preparation

The experiments presented here are conducted on Longmaxi shale at Sichuan basin, China. The block of full-size cores was sealed when received, and was subsequently stored in inert oil after removal of the seal to avoid desiccation effects.One shale sample perpendicular to beddings were prepared from nanoindentation and they were cut into small pieces of about 8 × 5 × 5 (length × width × thickness) mm, label by TA1, TA2, and TA3 respectively. Meanwhile, the TOC content was determined by a Leco CS230 carbon/sulfur analyzer (LECO Corporation, St. Joseph, MI, USA) on side cores. In detail, side cores were crushed to 100 mesh, treated with hydrochloric acid to remove carbonates and washed with distilled water. After being dried at 70 °C, the TOC values were measured as approximately 7.8 %, indicating shale samples are organic rich. For all cases, a surface RMS roughness lower than 70 nm (measured with the Berkovich tip) over an area of 80 μm × 80 μm was considered to be adequate for performing nanoindentation test. The machine Zeta-20 was used to randomly scan the surface and check the roughness. To provide reliable nanoindentation results, the top and bottom of surfaces were mechanical polished by SiC abrasive papers and polishing by argon ion milling is also necessary. The nanoindentation accuracy was initially calibrated using a fused silica individual calcite crystals, copper, aluminum and under a wide range of applied maximum loads from 0.1 mN to 500 mN standard before the nanoindentation test.

3.2. Nanoindentation design

The tests were conducted on the center of the sample with three 6 × 5 grid sets and 15 μm spacing. Each sample was indented 90 times in identical samples before and after SC-CO_2 treatment using a Berkovich pyramidal indenter (Agilent G200). The indentation apparatus has a displacement resolution of 0.01 nm and a load resolution of 100 nN during standard loading. Curves that display a kink in the loading curve, most likely due to grain displacement, were discarded as they will give an overestimated maximum depth and inferred contact area, and consequently an underestimated indentation moduli (Eq. (1)). A manual correction can though be considered. As for the CLR mode, the indenter was advanced at a rate of 2.0 mN/s to 500 mN, held at constant load for 15 s, and unloaded at 2.0 mN/s. and the maximum loading force was set as 500 mN, thus can make sure the measurement relates to lower indentation depth. Subsets of the data were analyzed to determine the effects of different hold times. The holding time of 15 s was applied to investigate the viscoelastic properties of shale. As for the CSM mode, the maximum indentation depth (h_{max}) was selected to be 3000 nm. The Continuous Stiffness Measurement (CSM) method was adopted using a constant target indentation strain rate (h/h) of 0.05 s^{-1}, where h is indentation depth. The three samples were performed by nanoindentation before and post SC-CO_2 fluid treatment. Fig. 2 presents a schematic diagram of the SC-CO_2 apparatus. This apparatus consists of a temperature control system, an experimental unit, and a supercharger. To control the experimental pressure and temperature conditions, the injection system shown in Fig. 2 was specially made for these experiments. The CO_2 was fed from a bomb to a cylinder of the syringe pump, which has a capacity of 400 mL. The volume of is reaction chamber is 150 ml. In order to fill the cylinder as full as possible, the CO_2 was cooled to a liquid state by circulating coolant in a cooling unit located above the cylinder. CO_2 becomes supercritical fluid above its critical temperature of 31.1℃ and critical pressure of 7.38 MPa, while it is liquid at lower temperatures. To simulate the reservoir pressure and temperature environment, the pressure and temperature is set to 45 MPa and 70℃, respectively.

3.3. Experimental process

We performed nanoindentation experiments over the surface of three shale samples after argon ion milling. To understand the fracture mechanisms and the relationship between the mechanical properties of shale and indentation fracture types, we studied the fracture surface morphology and mineral compositions changes through the combination of SEM and eEDS. The reaction chamber was injected into 20 ml distill water initially, which is able make all samples immerse within water. Because the injection volume of $SCCO_2$ is much larger than injected water volume, thus the water can be supersaturated with CO_2 (The maximum amount of CO_2 in volume is less than 0.03 mol) [30]. Afterward, all three shale samples were treated with SC-CO_2 and water environment for 48 h. The next step was to repeat the nanoindentation experiments under the same conditions while SEM and EDS were performed together. In particular, the SEM and EDS mapping in this study not only focused on the rock indented surface but also on the interior, while the relationships between different mineral compositions, mechanical properties and indenter fractures were also examined. The mechanical properties were calculated from the displacement vs. time curves. Fig. 3 shows the experimental scheme of this work.

4. Results

4.1. Indentation fracture morphology and mineral compositions

SEM and EDS were combined to observe the morphology and topography of the rock surfaces while analyzing the mineral compositions. According to SEM images, quartz usually appears as a flat light to medium gray mineral with somewhat rounded edges that displays undulatory extinction and generally has a smooth surface. Clays are often found in the matrix of shale samples and appear as very fine grained crystals with low interference color. As many clay minerals are micaceous, larger crystals can be identified by their crystal shape, e.g., platy and booklet euhedral kaolinite, fibrous illite, rosette-shaped aggregates of chlorite. The quartz or clay rich shales can be further identified by EDS mapping since each mineral has a specific chemical/element composition, e.g., Quartz is a hard, crystalline mineral composed of silicon and oxygen atoms, while Clay minerals are hydrous aluminium phyllosilicates, sometimes with variable amounts of iron, magnesium, alkali metals, alkaline earths, and other cations. Since the main minerals of the shale used in these experiments are quartz, clay and small amounts of feldspar, calcite and pyrite, with the help of the EDS technique, our shale samples can be divided into quartz-rich and clay-rich areas such that we can compare the nanoindentation fracture geometry differences between these areas. After the nanoindentation experiments, the indentation depth, mechanical properties are also combined together to double check measurement area belongs to quartz-rich area or clay rich area. In general, larger indentation depth and lower mechanical parameters are responses to clay rich area). Fig. 4(a) and (b) show SEM images of the nanoindentation grid matrix with and without SC-CO_2 treatment. It can be seen that the indentation imprint of shale with SC-CO_2 treatment is different from the imprint of shale without SC-CO_2 treatment while the surface of shale with SC-CO_2 treatment appears to be rougher. That's because the dissolution of CO2 in water is able to create the carbonic acid, thus partial minerals can be dissolve. In addition, organic matter extraction, and structural changes are also possible factors for surface morphology alteration [31,32]. Because SC-CO_2 is a good solvent for organic compounds such as benzene, toluene, ethyl-benzene, and xylene (BTEX); phenols; and polycyclic aromatic hydrocarbons (PAHs) [33], the kerogen in the micro- and nano-pores can be removed by SC-CO_2 fluid, creating more pore space. Because kerogen is also associated with the clay matrix, many pores can be created between clay crystals. The adsorption effect of SC-CO_2 on shale has been demonstrated by shale swelling in traditional triaxial mechanical testing [34,35]. The natural heterogeneity of rock is

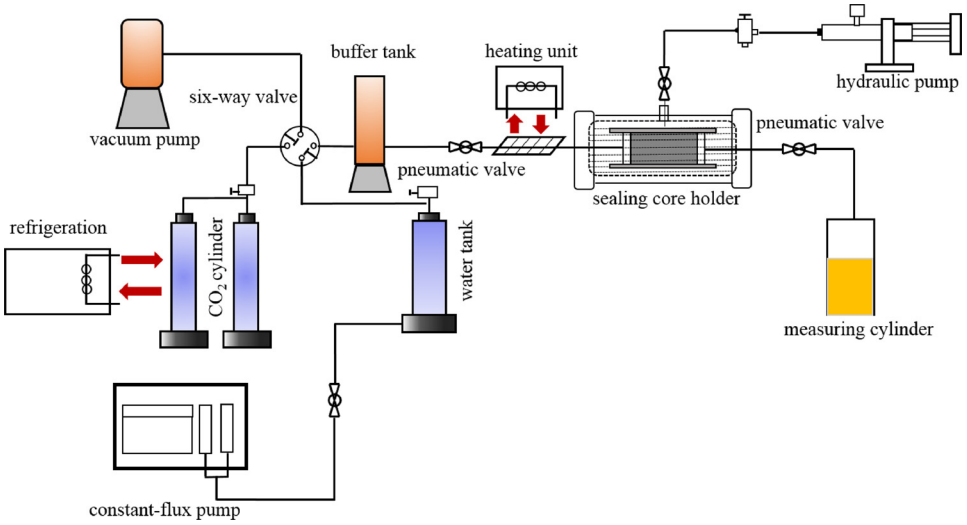


Fig. 2. Schematic diagram of the rock and SC-CO_2 interaction apparatus.

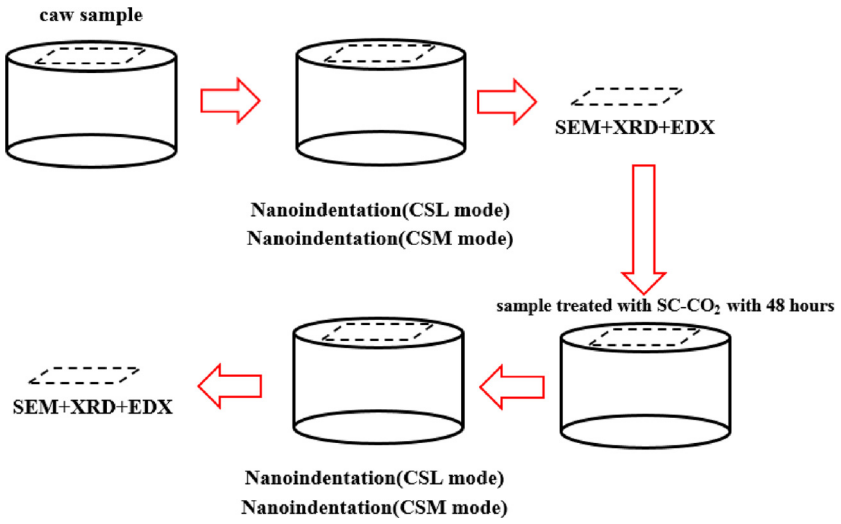
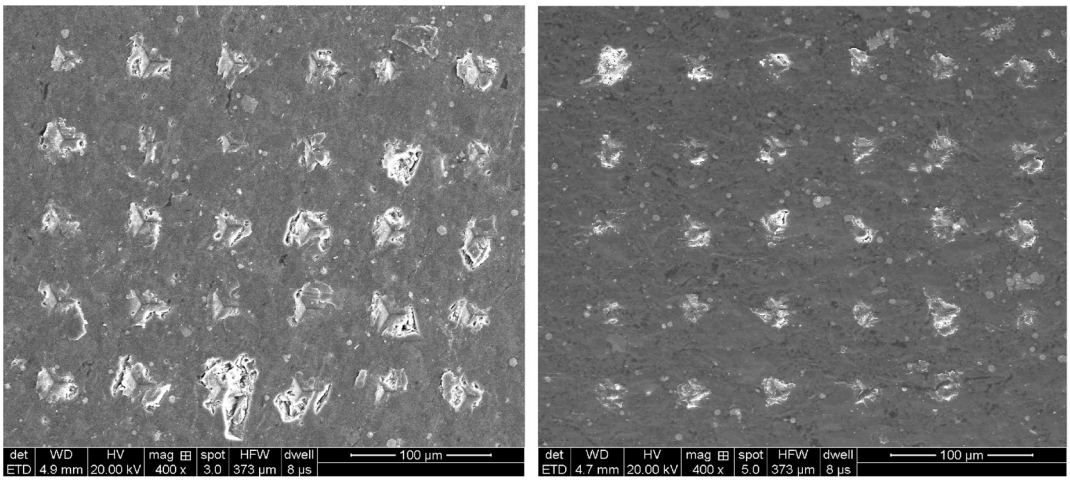
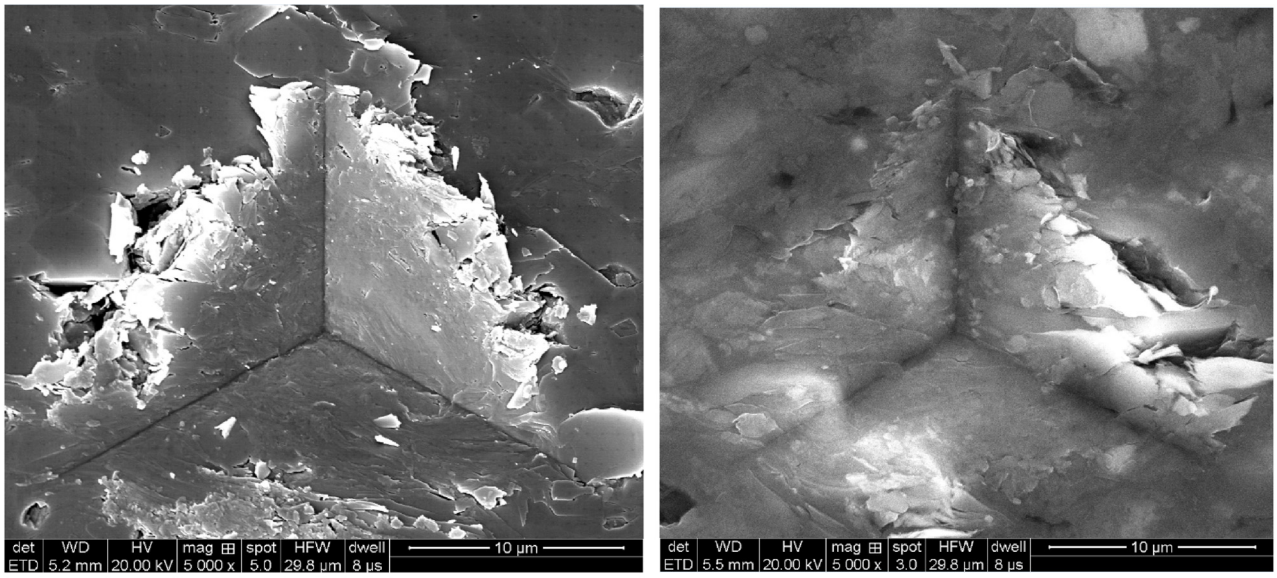


Fig. 3. Experimental design of the nanoindentation schematic diagrams for interactions between shale and SC-CO_2.

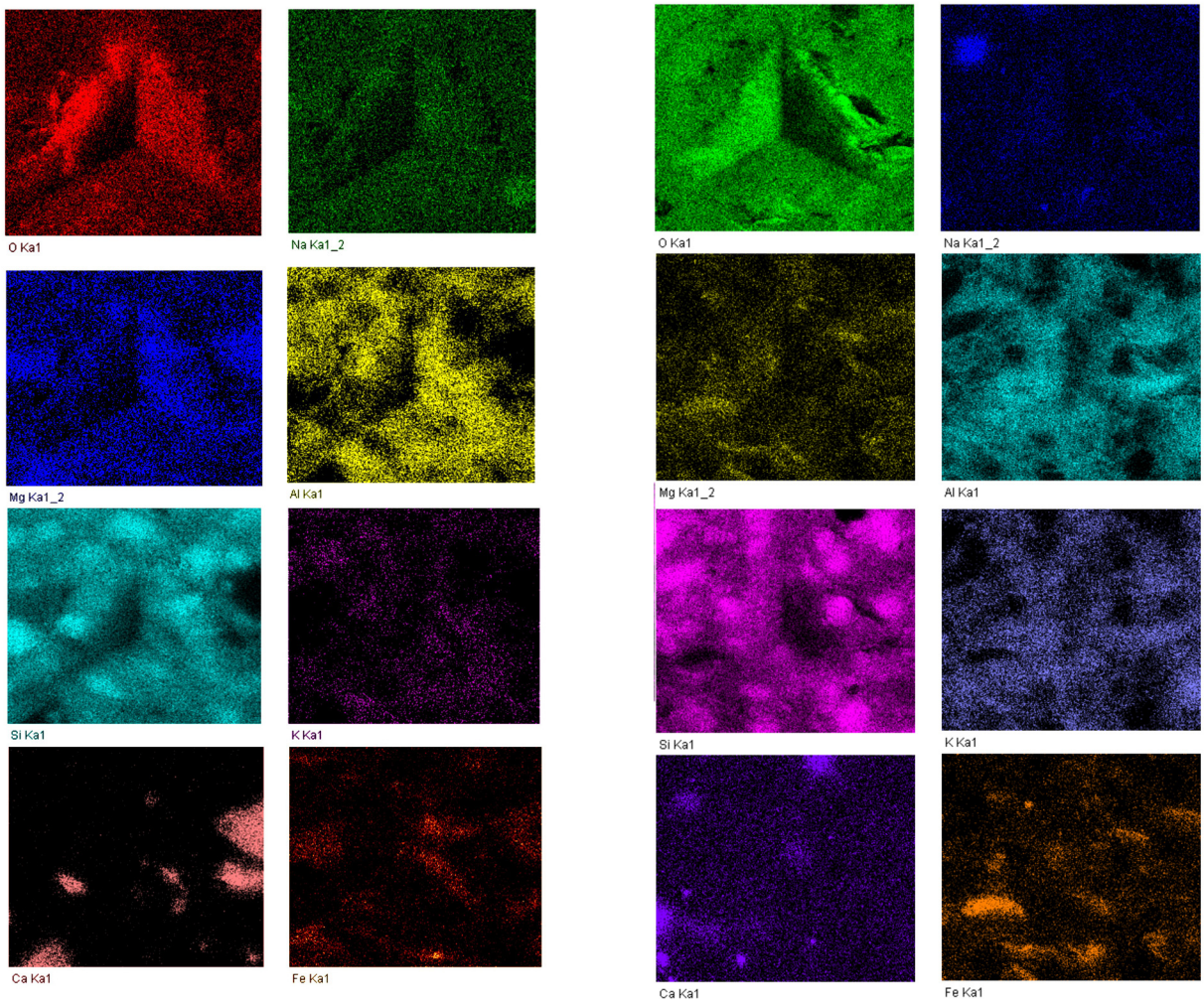


(a) postindentation region of the nanoindation sample before SCCO₂ treatment (b) postindentation region of the nanoindation sample after SCCO₂ treatment

Fig. 4. SEM photographs of the grid indentation matrix before and after SC-CO_2 treatment of the shale sample.

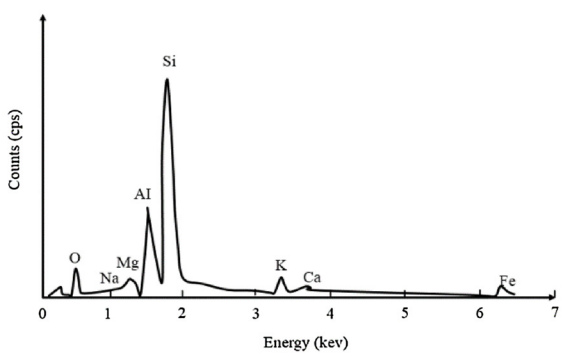


(a) post-indentated quartz-rich region before $SCCO_2$ treatment **(b) post-indentated quartz -rich region after $SCCO_2$ treatment**

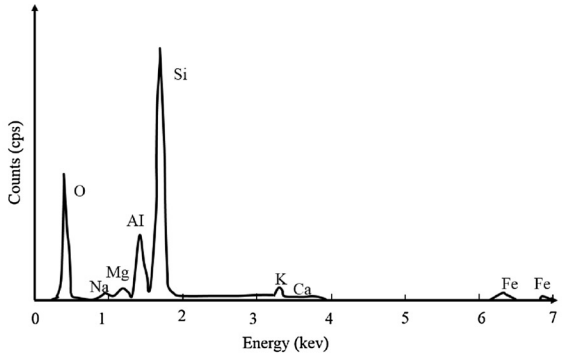


(c) elemetn distribution for quartz-rich region before $SCCO_2$ treatment (d) element distribution for quartz-rich region after $SCCO_2$ treatment

Fig. 5. SEM and EDS mapping photograph before SC-CO_2 treatment on shale.



(e) EDS mapping results of quartz-rich region without SC-CO_2 treatment



(f) EDS mapping results of quartz-rich region with SC-CO_2 treatment

Fig. 5. (*continued*)

regarded as a main factor behind the formation of multiple cracks during nanoindentation, and the fracture propagation in the post-treated shale is primarily controlled by the mechanical property and brittleness degradation of rock by SC-CO2 treatment. For example, the rock structure has been weakened, as shown by the irregular grain margins. Certainly, because limited test area, local and minor mineral or pore structure changes could affect nanoindentation results even if the dissolution volume of mineral and kerogen might be quite small.

Figs. 5 and 6 display the SEM and EDS images of single indentations on the quartz-rich and clay-rich regions before and after SC-CO_2 treatment. The close examination of a single indentation fracture surface indicates the changes in the interior structures and indentation fracture types. It shows that the fracture propagation imprint of shale with SC-CO_2 treatment is more obvious than that of the raw shale sample, and the three individual fracture widths from each indentation in shale with SC-CO_2 treatment are larger than that in the raw shale sample. In addition, the fracture propagation path varies within a local area, regardless of whether it occurs in a quartz-rich or clay-rich area. More importantly, we can see more micro-factures after SC-CO_2 treatment in both clay-rich and quartz-rich areas, which is because the rock heterogeneity increases and the mineral bonding strength decreases with treatment, thus resulting in microfractures and curve fracture extension.

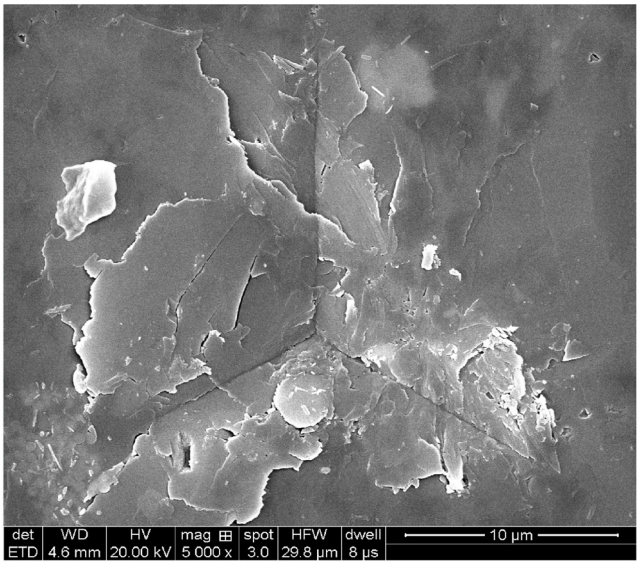
The chemical compositions of shale with and without SC-CO_2 are shown in Fig. 5(c) and (d). The presence of both Si and O elements can be used to find quartz e.g. the bright spots in the Si element map image. Furthermore, quartz is indicated by quartz grain morphology and even surface. K and Na-feldspar, orthoclase and albite (bright spots in the lower middle and upper right areas in the K element map and bright spots in Na element map) can be identified by the existence of K, Si and O and the trace of Al. In general, illite has lath-shaped to fibrous shape. Ca can be used to find carbonate minerals and clay minerals (bright spots in the Ca element map). Moreover, the other elements e.g. iron (Fe), sodium (Na), magnesium (Mg), potassium (K) and calcium (Ca) are sporadically distributed. Mg and Fe peaks in the EDS images could appear in these areas as a minor carbonate cements of dolomite or ankerite may develop from diagenesis. More importantly, ions can be released from the initial rock minerals into the solution during the SC-CO_2 and shale reactions, leading to the dissolution and new precipitation of the minerals. Comparison of element before and post SC-CO_2 treatment (Fig. 6(c) and (d)) reveals that there is little change in the concentration of Si, K, Mg and AI, but the concentration of O was almost doubled In general, the dissolution of K-feldspar and clay minerals (kaolinite, illite and illite/smectite mixed layers) can release K, Al, Fe, Mg and other metal ions, but the SC-CO_2 has little effect on quartz. The decrease of Ca element concentration indicates the dissolution of a certain content of calcite and feldspar.

4.2. Load-displacement curves

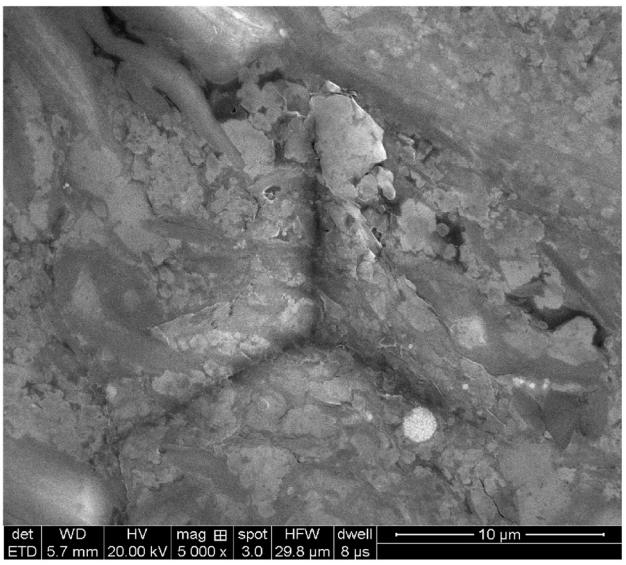
Fig. 7(a) and (b) show the load-displacement curves from the CSM mode for shale before and after SC-CO_2 treatment in the quartz-rich region and clay-rich region. It was found that the maximum load was higher in the untreated rock sample than in the treated rock sample, which means that the indentation needed a smaller load to reach the predetermined indentation depth in the raw shale sample. Thus, the mechanical properties of the shale sample decreased after interaction between the SC-CO_2 and the rock minerals according to the CSM results. Fig. 8(a) and (b) display a load-displacement comparison of shale before and after SC-CO_2 treatment in the quartz-rich region and clay-rich region. Since the maximum load needs to be predetermined under the CLR mode, deeper indentation depths generally reflect lower mechanical properties of a material. Comparison of the maximum indentation depth reflects that the values of mechanical properties are higher in the untreated rock sample than in the SC-CO_2 treated rock sample, although the maximum load is identical for all tests in CLR mode. According to the nanoindentation results, it can also be found that the post-SC-CO_2 rock sample showed an increased maximum indentation depth in both the quartz-rich region and the clay-rich region, which is consistent with the results extracted from CSM mode. The greater indentation penetration depth signified that a greater contract area and smaller hardness and Young's modulus values were calculated for the SC-CO_2-treated rock sample. Thus, the rock mechanical properties can be expected to decrease after interaction between the shale and SC-CO_2. However, even if both the clay-rich region and the quartz-rich region suffer from SC-CO_2, the effects of SC-CO_2 on the load vs. displacement curve are different. In particular, compared to the nanoindentation results obtained under CLR mode, a more obvious difference can be easily observed under CSM mode. More information can be found in that clay-rich regions seem be influenced more greatly than are quartz-rich regions, an effect which is primary due to clay adsorption by SC-CO_2.

4.3. Effect of CO_2 on the mechanical characteristics of shales under CSM mode

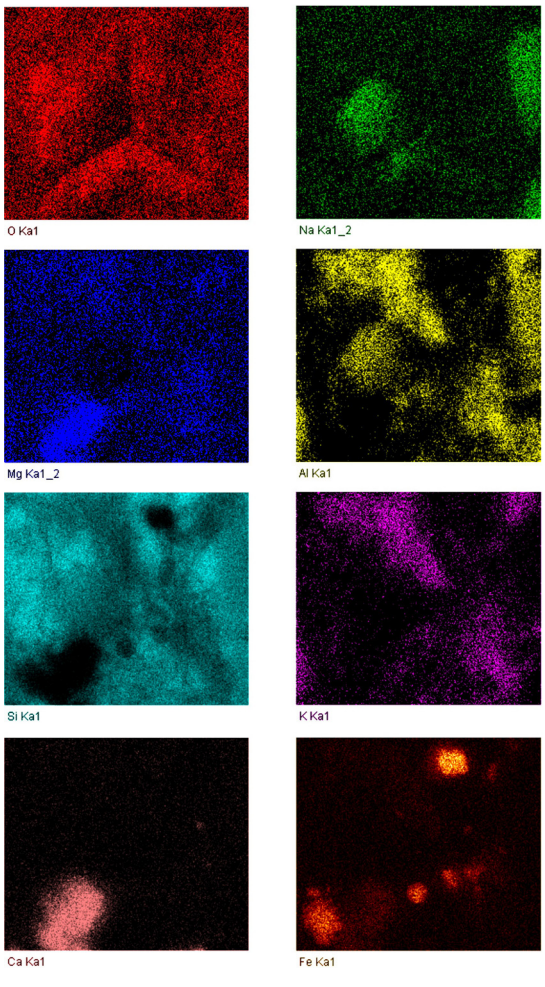
Fig. 9 shows the typical mechanical property changes measured by CSM tests on an untreated shale sample and a SC-CO_2-treated shale sample. Although the results show that the contact stiffness, elastic modulus and hardness are functions of the contact depth, the patterns of the mechanical properties curve relative to displacement are different under CSM mode for quartz-rich regions and clay-rich regions. The change of the contact stiffness can be used to reflect the homogeneous rock difference from the surface to the rock interior. Thus, although the contact stiffness values of both the SC-CO_2-treated shale sample and the untreated shale sample increase with a decreasing slope, the scattering of contact stiffness is more obvious for the clay-rich



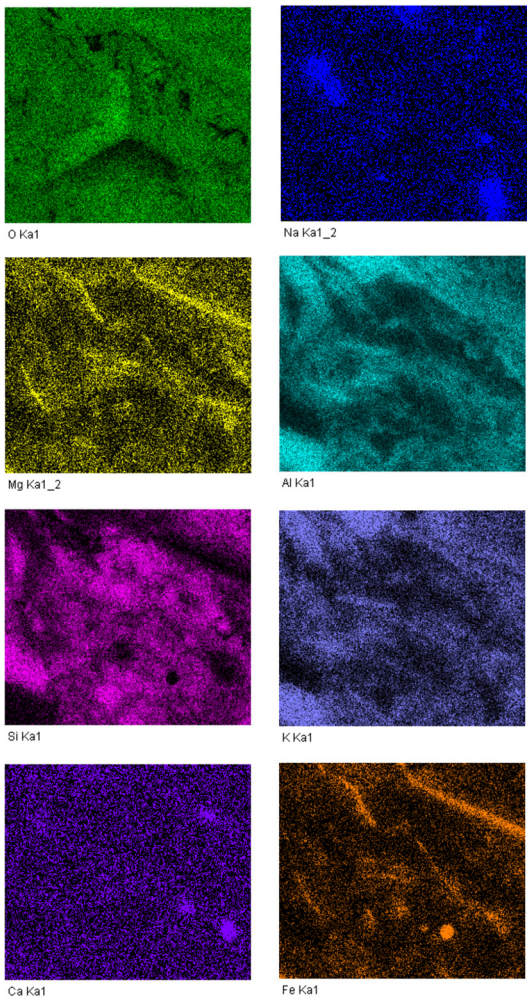
(a) post-indentated clay-rich region before $SCCO_2$ treatment



(b) post-indentated clay-rich region mineral after $SCCO_2$ treatment



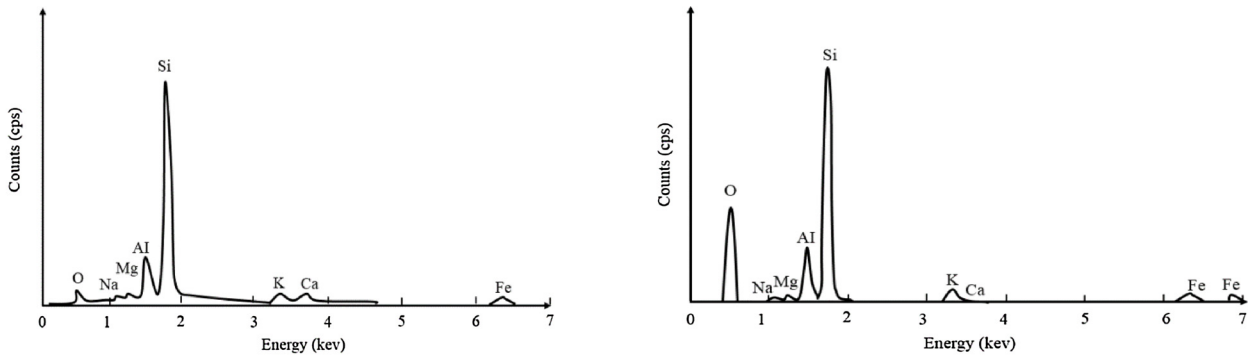
(c) EDS mapping results of clay-rich region before SC-CO_2 treatment



(d) EDS mapping results of clay-rich region after SC-CO_2 treatment

Fig. 6. SEM and EDS mapping photograph after SC-CO_2 treatment on shale.

region than for the quartz-rich region, especially for the raw shale sample, because more nanopores might exist within the clay-rich region. The results of these experiments indicate that the difference in contact stiffness is not evident between the untreated rock sample and the SC-CO_2-treated rock sample for the quartz-rich region, whereas a clear difference is shown between the untreated rock sample and the SC-CO_2-treated rock sample for the clay-rich region, which also provides evidence for surface heterogeneity between the quartz-rich region



(e) EDS mapping results of clay-rich region without SC-CO_2 treatment **(f) EDS mapping results of clay-rich region with SC-CO_2 treatment**

Fig. 6. (*continued*)

and clay-rich region. Moreover, the stiffness data is slightly more scattered at deeper penetration depths, because the mineral particle distributions are more even on the surface without external loading due to mechanical and argon ion polishing, while the bottom below the surface is more homogeneous. The increasing load results in rock deformation and increases the heterogeneity; thus, contact stiffness data is more scattered with increasing penetration depth.

To further discuss the mechanical properties of shale before and after SC-CO_2 treatment, the typical elastic modulus and hardness changes with increasing indentation depth are compared in Fig. 9(b) and (c). The hardness and Young's modulus values of quartz-rich regions commonly followed a decreasing tendency and then approached a stable value, whereas the hardness and Young's modulus values of clay-rich regions increased initially and then fell slowly to a stable value. In other words, *E* and *H* reveal significant variations with increasing penetration depth for the given case. The drops in hardness and elastic modulus were primarily due to mechanical strength degradation and the occurrence of fractures and defects, a finding which is also consisted with the load vs. displacement curve. Moreover, another important result extracted is that the elastic modulus of the SC-CO_2 = -treated rock sample in the clay-rich regions decreased more obviously than did that in the quartz-rich regions, while the SC-CO_2-treated rock sample had almost the same hardness in quartz-rich regions and clay-rich regions. The difference was mainly because the Young's modulus calculation used the indentation surface area and contact stiffness as inputs, but the hardness computation only depends on the indentation surface area. The decrease of contact stiffness in clay-rich regions is the reason for the evident decrease in the Young's modulus.

It can also be observed that the elastic modulus (*E*) and hardness (*H*) both decreased with increasing indentation depth (*h*), whereas they gradually approached stable values in CSM mode. Great fluctuation of the curve pattern can be observed for the elastic modulus and hardness values. The penetration depth of 1000 nm is possibly the greatest reaction distance for this sample, which means that the SC-CO_2 has an evident influence on the mechanical properties from the surface to this depth, depending on the SC-CO2 seepage capability. The stable values of *E* and *H*below this depth are generally considered the average elastic modulus and hardness of the shale matrix. To avoid possible indentation effects on the mechanical results such as the sink-in phenomenon, 1500 nm was set as the effective dentation depth for mechanical properties calculations in all indentation tests. The Young's modulus and hardness were averaged for each indentation.

The comparison of the average Young's modulus and hardness values under CSM mode before and after SC-CO_2 treatment is presented in Fig. 10. It can be concluded that the bulk Young's modulus decreased from 52.14 GPa, 52.63 GPa and 46.91 GPa to 45.77 GPa, 46.18 GPa, and 44.25 GPa for TA1, TA2 and TA3, respectively. The bulk hardness decreased from 3.26 to 2.21 GPa, from 3.21 to 2.18 GPa and from 3.18 to 2.17 GPa for samples TA1, TA2 and TA3, respectively. According to the results, there were apparently 32.2 %, 32.0 % and 31.7 % drops in the hardness for the three samples, TA1, TA2 and TA3, respectively. The average Young's modulus decline was at the proportion of 12.2 %, 12.2 % and 10.8 % for samples TA1, TA2 and TA3, respectively. Thus, the average drop rates of the three samples were 32.01 % and 11.75 %

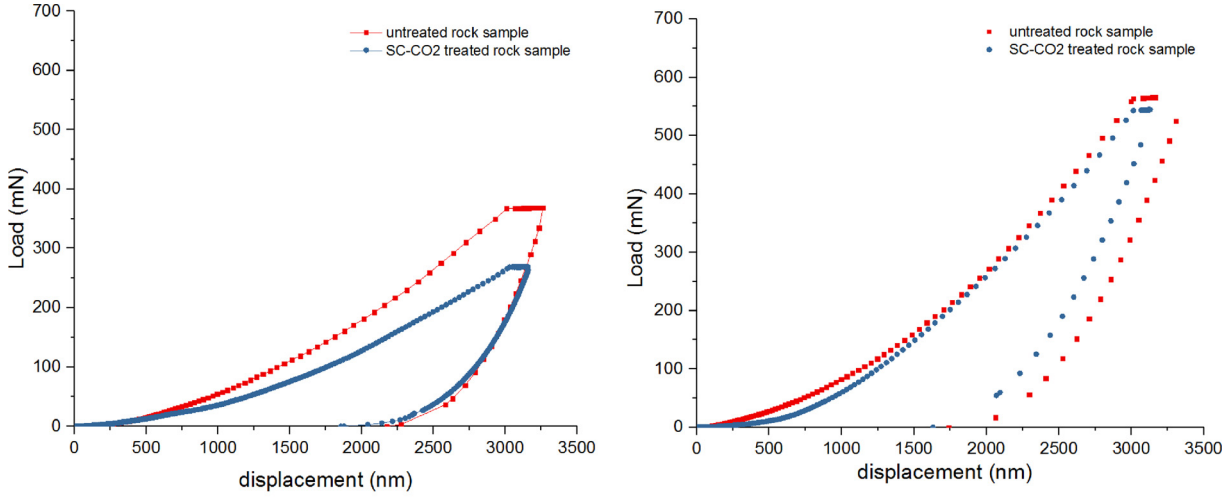


Fig. 7. Loading and unloading curves for shale with and without SC-CO_2 treatment under CSM mode.

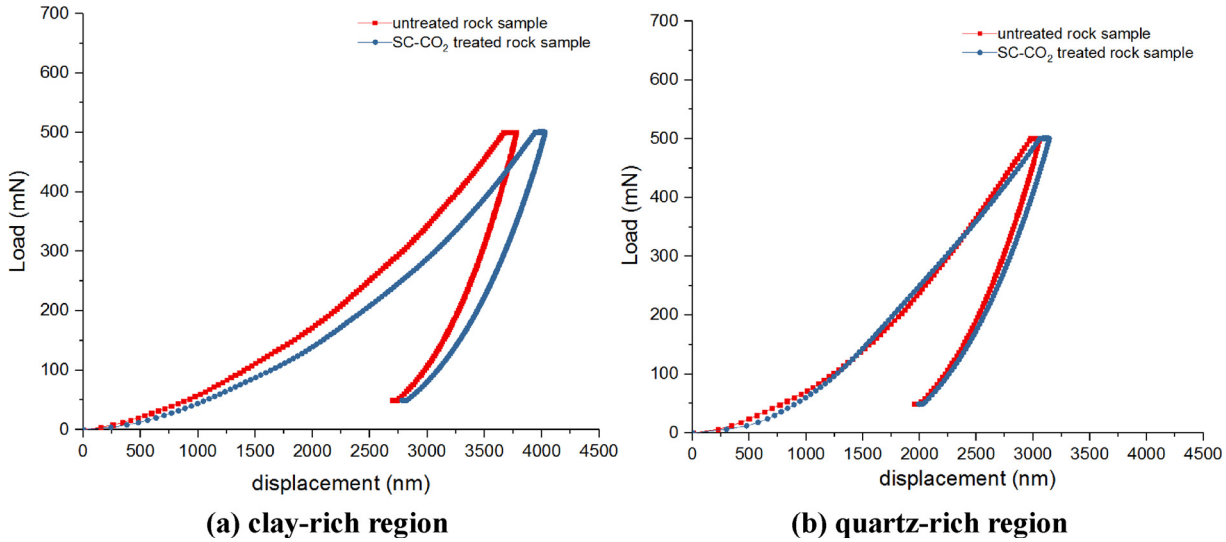


Fig. 8. Loading and unloading curves for shale with and without SC-CO_2 treatment under different CLR mode.

for hardness and Young's modulus, respectively.

4.4. Effect of SC-CO_2 on the mechanical properties of under CLR mode

Nanoindentation by CLR mode was used to investigate the mechanical properties changes before and after SC-CO_2 treatment at the surface and under certain depths in the samples. In addition to the discussion of Young's modulus and harness under CSM mode, the fracture toughness can be further investigated. The Young's modulus, harness and fracture toughness values measured before and after SC-CO2 treatment for TA1 are shown in Fig. 11. Data are shown to fluctuate for each mechanical properties on individual indentations. The indentation moduli have a wide range from 35.52 GPa to 65.09 GPa. The hardness have a wide range from 1.86 GPa to 4.97 GPa. And the fracture toughness have a wide range from 4.69 MPa $m^{0.5}$ to l 8.76 MPa $m^{0.5}$. The heterogeneity of the sample can explain the scattering of the data. Thus, it's necessary to include a high enough number of indentations for the mechanical properties calculations through the averaging computation method.

The comparison of the average Young's modulus, harness and fracture toughness values for the three samples extracted from nanoindentation data are shown in Fig. 12(a)–(c). It can be concluded that the bulk Young's modulus decreased from 49.24 GPa, 50.63 GPa and 47.01 GPa to 43.56 GPa, 44.04 GPa, and 42.94 GPa for TA1, TA2 and TA3, respectively. The bulk hardness decreased from 3.19 to 2.25 GPa, from 3.09 to 2.18 GPa and from 3.05 to 2.14 GPa for samples TA1, TA2 and TA3, respectively. The bulk fracture toughness decreased from 6.67 to 5.94 GPa, from 6.87 to 5.87 GPa and from 6.38 to 5.85 GPa for samples TA1, TA2 and TA3, respectively. According to the results, there were apparently 29.46 %, 29.44 % and 29.83 % drops hardness for the three samples, TA1, TA2 and TA3, respectively. The average Young's modulus declines were in the proportion of 11.35 %, 13.01 % and 8.77 % for the TA1, TA2 and TA3 samples, respectively. The fracture toughness showed a 10.94 %, 14.56 % and 8.31 % drop for the TA1, TA2 and TA3 samples, respectively. The average drop rate of the three sample was 11 %, 29.5 %, and 11.3 % for the hardness, Young's modulus and fracture toughness values, respectively.

5. Discussion

The indentation-induced cracking pattern depends on the indentation load and the indenter geometry. The comparison of the raw shale sample clay-rich and quartz-rich regions indicates two different possible fracture mechanisms. For clay-rich regions, the number of fractures in the middle of the indentation region is relatively small because strong load energy accumulates at the indentation tip, and micro-fractures possibly proceed from open conditions to compaction conditions. In quartz-rich regions, more micro-fractures can be generated because the mechanical strength of quartz grain is much higher than that of soft clay; thus, the indentation fracture prefers to propagate along the quartz grain. Under the same conditions, indentation fractures more easily appear at intragrain boundaries and in areas where there exists a weak strength zone. After SC-CO_2 interaction, we can see more pores of bigger sizes and a softer mechanical strength; thus, multiple fractures are easier to create with the existence of defects and micro-fractures.

The mechanical properties degradation expected to occur after interaction between SC-CO_2 and rock minerals in organic shales was examined using different nanoindentation experimental designs. These two methods generated similar results under both CSM mode and CLR mode because the indentation depth was approximately 3000 nm. However, upon comparing nanoindentation results obtained under CSM and CLR mode, the SC-CO_2 exerts a clearer influence on the load vs. displacement curve, rock hardness and Young's modulus in clay-rich regions than in quartz-rich region. The reasons for such findings can be explained by the influence of SC-CO_2 on shale being mainly clay adsorption and mineral dissolution, but this explanation might only be applicable to shale with high clay and kerogen contents. Concerning the mechanical strength in quart-rich regions, calcite and feldspar dissolution should be the critical factor. The emergence of Fe precipitations possibly comes from silicate minerals, pyrite or ankerite. The Fe precipitations generally have larger mechanical strength, thus can contradict the weaken effect due to SC-CO2 treatment [36]. Additionally, there is a direct relationship between the mechanical properties decrease and the pore structure, experimental pressure and temperature of shale. The decline of the mechanical properties produces competing effects on shale gas stimulation and recovery. The injection of SC-CO_2 can reduce rock fracture toughness. In general, fracture toughness indicates a material's fracture capability when it is subjected to an external stress impact. The decrease of the fracture toughness reflects a low fracture energy required for initiation, which is beneficial for lower fracture breakdown pressure and complex fracture network generation. Furthermore, because of the different adsorption and kerogen removal capabilities for different mineral compositions in diverse areas, the adsorption-induced anisotropic and heterogeneous stresses can produce more defects and micro-fractures in shale, thus further decreasing the fracture propagation energy. As evident from the hardness and Young's modulus values obtained from the different

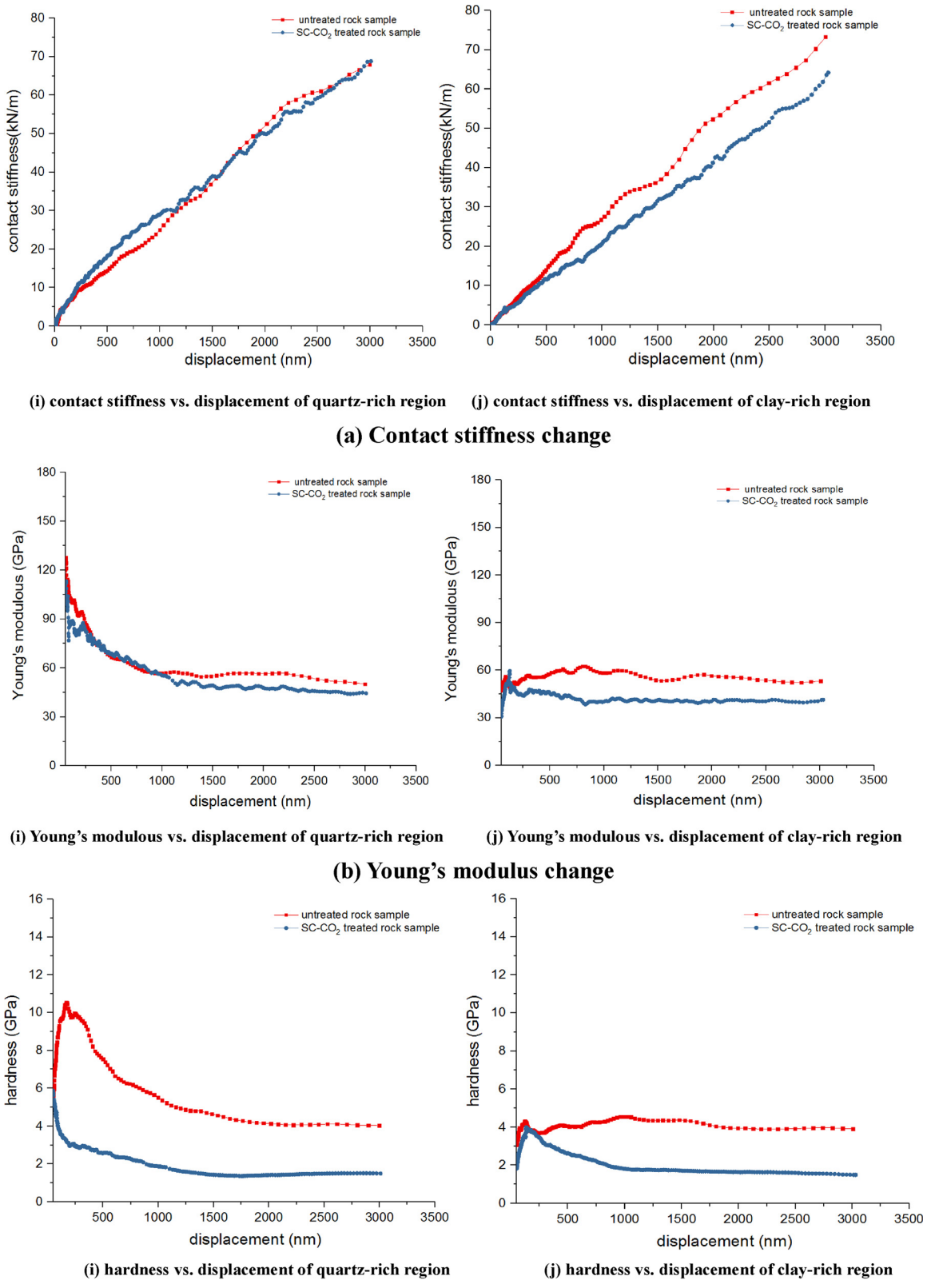
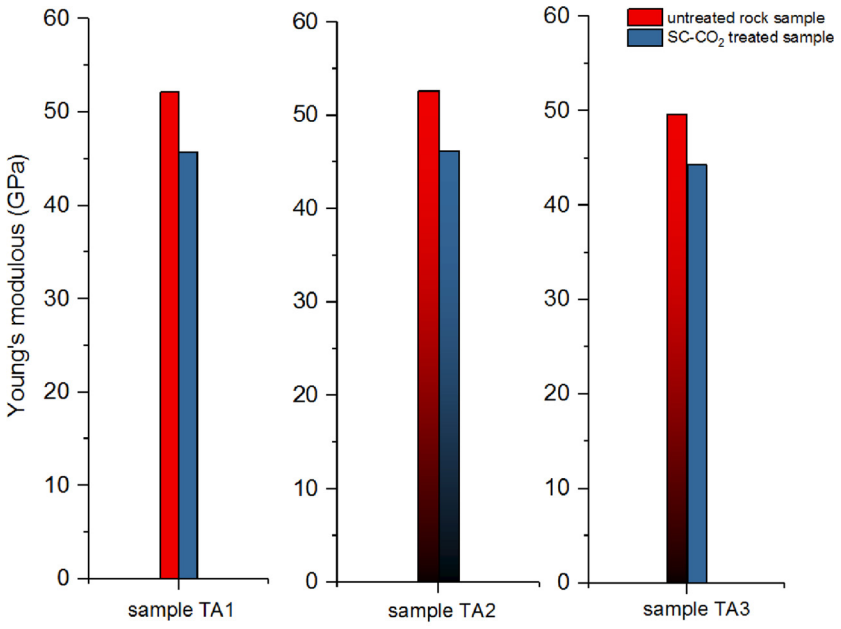
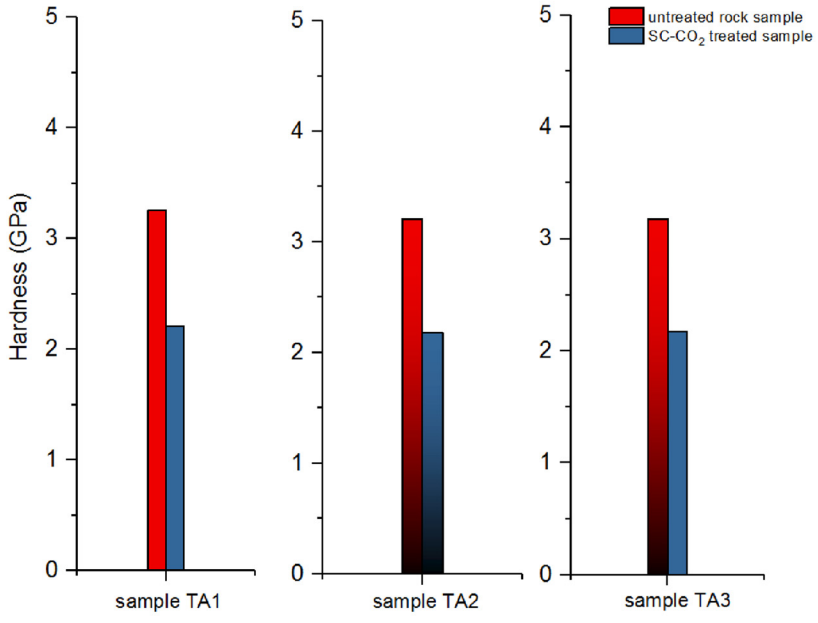


Fig. 9. Mechanical changes of shale with and without SC-CO_2 treatment under CSM mode.

nanoindentation tests under CSM mode and CLR mode, these two mechanical values also decreased by a large proportion. Thus, such sharp shale mechanical strength declines can increase the possibility for proppant embedment. Generally, if the viscoelastic properties of shale are considered, proppant embedment problems tend to be more severe in terms of sustaining fracture conductivity for long-term gas production [37–39]. Therefore, it's necessary to select good timing to inject the proppant before fracture closure during SC-CO_2 fracturing.



(a) Young's modulus



(b) hardness

Fig. 10. Average mechanical properties changes before and after SC-CO_2 treatment under CSM mode.

6. Conclusions

In this paper, the continuous stiffness measurement (CSM) mode and constant loading rate (CLR) nanoindentation method was successfully applied for evaluating changes in the mechanical properties before and after SC-CO2 treatment. The experimental results show that the method can be effectively applied to characterize the shale mechanical properties at nanoscale, which is beneficial for overall geo-mechanical response estimation. In addition, the results can also provide more insights for determination of failure initiation position after SC-CO2 treatment

(1) Compared with the traditional CLR mode, we can measure the effect of SC-CO_2 from the rock surface to the rock interior with the CSM technique. The CSM and CLR methods provide similar nanoindentation results through the averaging method, which demonstrates the accuracy of these two approaches. Rock heterogeneous and anisotropic properties are the basic reasons behind the scattering of data during nanoindentation. In addition, with the aid of the EDS mapping technique, we can identify the mineral distributions of individual indentations, and the mechanical properties differences between clay-rich regions and quartz-rich areas can be compared.

(2) The difference between mechanical properties before and after SC-CO_2 treatment further demonstrates that the SC-CO_2-treated sample has a deeper maximum penetration depth than does the SC-CO_2-

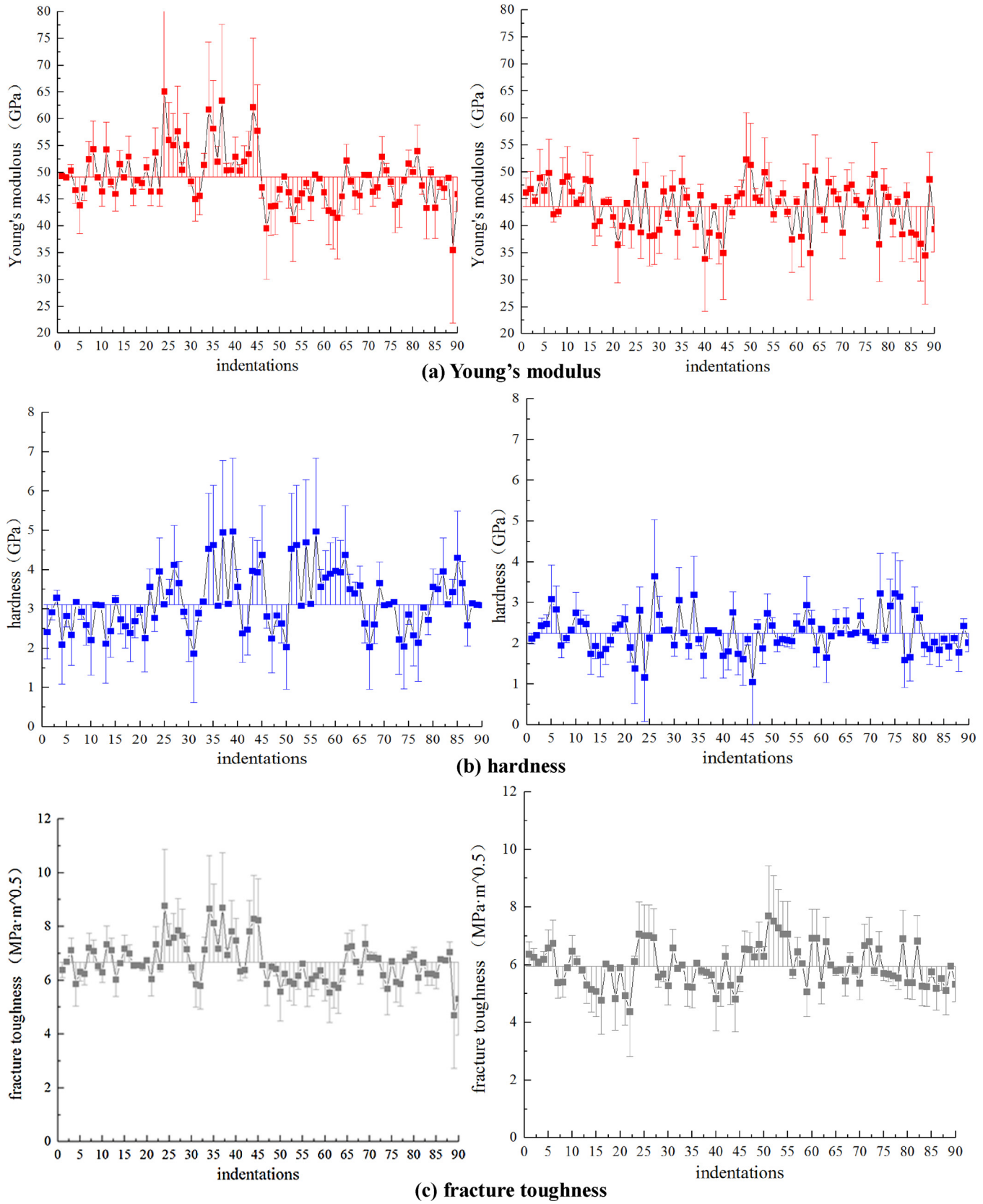


Fig. 11. Mechanical properties changes before and after shale and SC-CO_2 interaction for the grid matrix of TA1.

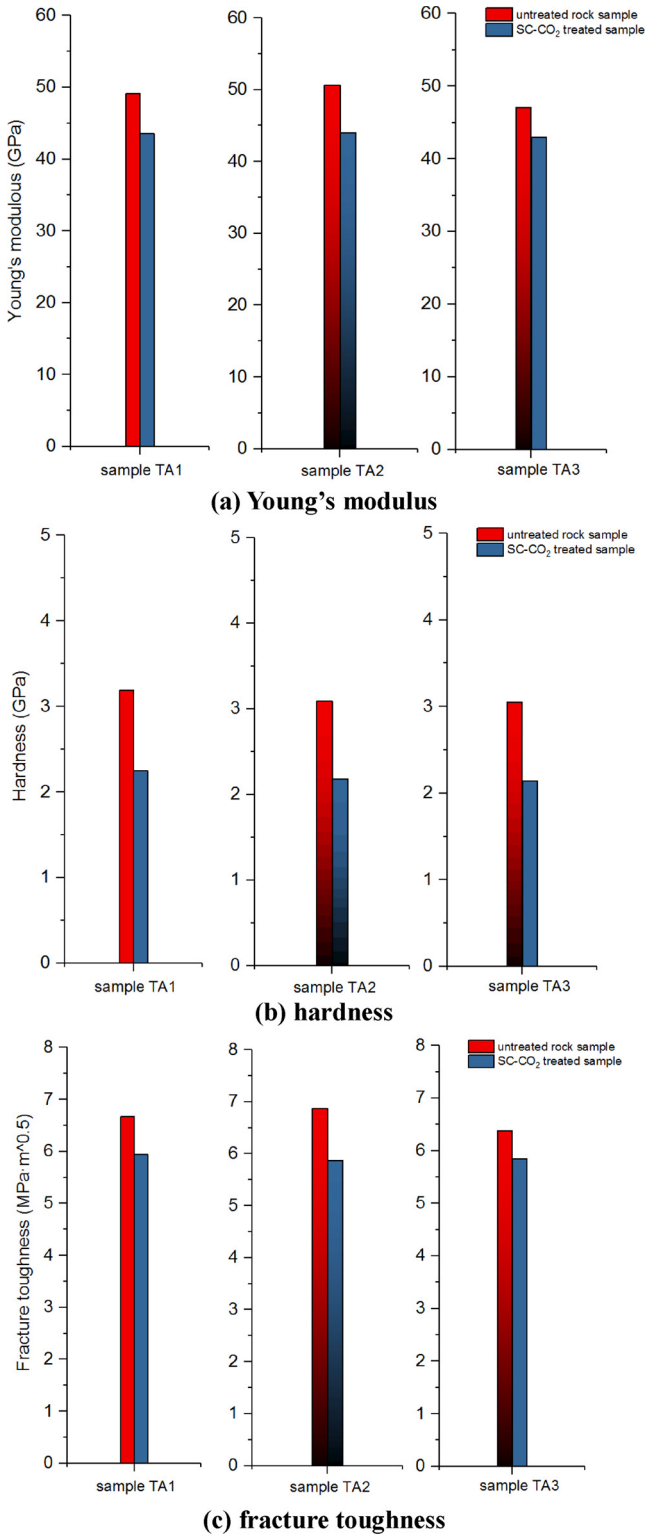


Fig. 12. Average mechanical properties changes before and after SC-CO_2 treatment under CLR mode.

untreated sample; thus, a greater contact area for indentation is expected, which is consistent with the SEM photograph. Moreover, the treatment of shale with SC-CO_2 can result in distinguishable and quite significant mechanical properties losses. The average rates of decrease of the three samples were 32.01 % and 11.75 % for the hardness and Young's modulus values under CSM mode, while the rates of decrease for the three samples were averaged at 29.5 %, 11.0 %, and 11.3 % for the hardness, Young's modulus and fracture toughness values, respectively, under CLR mode. The comparison of the mechanical properties of areas with different mineralogies illustrates that clay-rich areas are more easily subjected to mechanical properties losses and structure alternations through SC-CO_2 treatment, and large declines in the maximum load and mechanical strength were observed in CSM mode, along with a pore size increase and the presence residual voids. However, less change in the rock mechanical strength and pore structure was observed in the quartz-rich area after SC-CO_2 treatment. The dissolution of CO_2 in water is able to create the carbonic acid while ions from the initial rock minerals were released into the solution during the reactions, leading to the dissolution and precipitation of the minerals. Therefore, the decline in mechanical properties and changes in pore structure correlate well with the mineralogical composition. Moreover, the reason behind the mechanical properties decrease in clay-rich regions is primarily CO_2 adsorption, whereas minor mineral dissolution governs the considerably smaller decline of the mechanical properties in quartz-rich regions.

(3) With the help of SEM analysis, we further discussed the fracture mechanisms before and after SC-CO_2 treatment. Multiple microfractures and voids could be observed around the indentation fractures in SC-CO_2-treated shale, and only one manifest indentation fracture could be observed in raw shale. The increase of microfractures and the decline of rock fracture toughness are beneficial for rock deformation and failures. In particular, the application of SC-CO_2 as a fracturing fluid can reduce the initiation pressure in shale gas stimulation. The decline of the mechanical properties will also bring some problems, such as potential proppant embedment and conductivity loss due to the sharp decline of the mechanical strength in long-term gas production. Therefore, the timing of proppant injection needs to be carefully considered.

The results presented in this paper can be used as a reference for brine-SCCO_2-shale interaction investigations under different conditions at the nano scale. Moreover, the fundamental theory about the kinetics and thermodynamics of the reactions between brine-SCCO_2-shale will be investigated in our future experiments.

Declaration of Competing Interest

The authors declare no conflicts of interest in relation to this work.

Acknowledgments

This work had been financially supported by the National Natural Science Foundation of China (51704324, U1762213, 41728004), National Science and Technology Major Project of the Ministry of Science and Technology of China (2016ZX05061), National Natural Science Foundation of Shandong Province (ZR2016EL07), and the Fundamental Research Funds for the Central Universities (17CX02040A).

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